

fungi

Date

- living organism, not bacteria or viruses
- cell → unit of life

fungi helps plants (roots)?

fungi ↔ bacteria are anti

antibiotic have fungi

bacteria & fungi fight each other

dark + moist = fungi

mucormycosis?

black fungus → some good for gut
→ black fungus during covid

fungi also have DNA --

bacteria and fungi has very good resistance.

Antibiotics - paracetamol



life of
bacteria

46 1960s → antibiotics came

plant's fungicides

John Naisbitt → We are drowning in info
but starved for knowledge.

cell cell cell
fungus is like human, bacteria is
not like human cell.

So making anti-fungal pill is difficult.

fungus - genomics - lab / candida - auris

models → trained to study DNA of
fungi
(Bio python, NLTK)

AFR Base

SEVA , V21 RISE 2024/25

→ Anti fungal resistance database

geography & fungi

susceptible and resistant

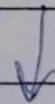
Recognising resistance → first step to its elimination

Shiny?

model → tells susceptible or resistant.

Where is database?
how fungi spreads?

Blood → DNA → Sequence
with of DNA
fungus



ML model

↙ ↘
Susceptible Resistance

did fungus & virus react?
~~what~~ ~~and~~ how actually fungi takes over body?

immunity —

fungal disease → CDC website.

fungal AMR?

One health in action

zoonosis (animal → human)



60% of emerging
infectious diseases
are zoonosis

3/4 human disease start from ~~#~~
animal.

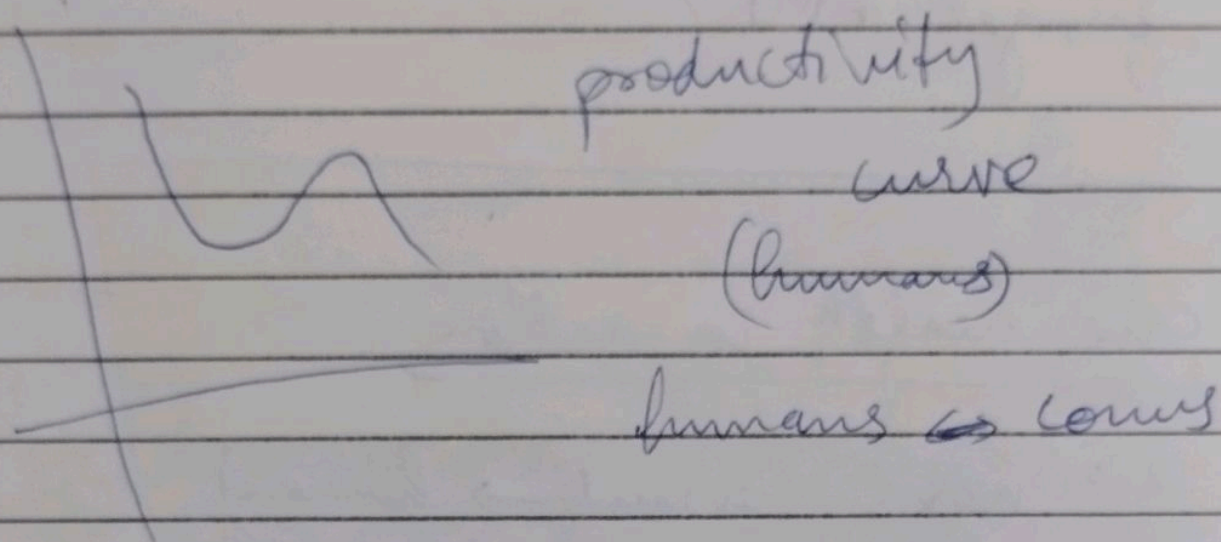
why do ~~B~~ we boil milk?

foot ~~in~~ mouth? from cow

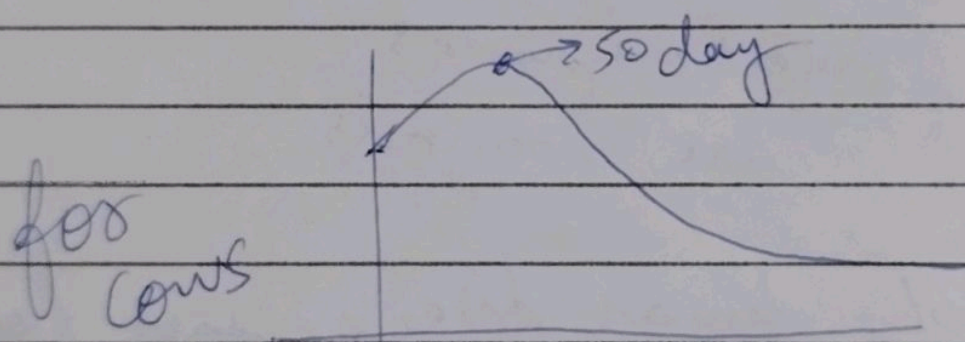
Syndromes ↔ productivity ↓



non-clinical
disease signs



305 days of milking



lactation curve

low productivity = increased work load

" milk yield = income disease

Syndromic, surveillance, signal.

Tatoos

Dermis - 2nd layer - tattoo ink.

Macrophages → dermal macrophages

↓
immune cells

↓
fights tattoo ink.

↓
eats ~~and~~ ink but can't break it
∴ tattoo healing time.

↓
die 8-10 days, then new macrophages take in the ink

Bone marrow transplant

→ nothing happens to tattoos

Keratinocytes

Fibroblasts

Macrophages

3D - Skin model

tattoo ink → Aqua, Propylene glycol, benzyl alcohol

ink goes beyond dermis → immunity response high

lymph, liver, blood $\rightarrow$ but can go

~~to~~ tattoo + lead in MRI machine
 $\rightarrow$ safe, but some studies
say opposite (4 days after
tattoo - 3 week to
heal)

lymphoma $\rightarrow$ cancer

Radiotherapy $\rightarrow$ tattoo

$\otimes$ tracking gut movement - tattoo

vaccination

laser removal.

Q-switched laser

cream + laser - in research.

A tale of drunken knots

Overlaps in Knot — in 2D projection

topoisomerase enzymes

entropy - cards

drunk	ant,	man,	bird
	↓	↓	↓
	1D	2D	3D

theorem →	↓	↓
	will get home	not get home

Random walk → dust particles,
stock market

DNA knots

Some knots attract, some repel
→ entropy increasing or decreasing

Galactic fossils $\rightarrow$ stars

" Archaeologist

~~Chemist~~ Star light

Studying stars:

$\rightarrow$ Chemistry

$\rightarrow$ Kinematics

$\rightarrow$ positions

star $\leftrightarrow$ dosa

photosphere -
visible surface

crispy outer layer

Hot core

spicy filling

unique recipe
of elements inside

blend of
spices inside

different stars

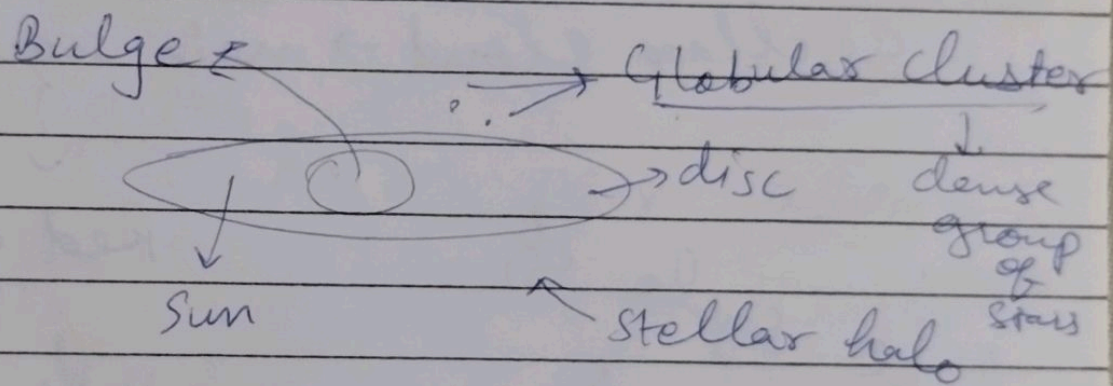
$\rightarrow$ population I $\rightarrow$ 3rd gen, newly born

$\rightarrow$ " II

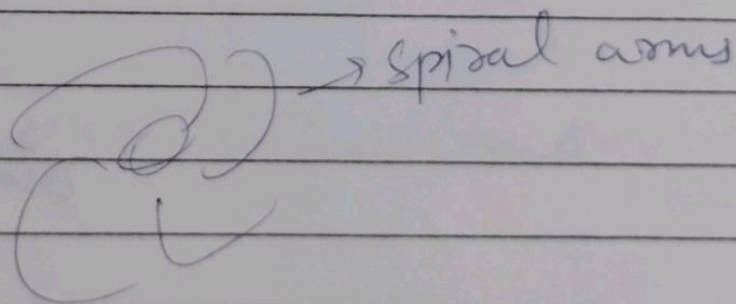
$\rightarrow$ " III

population I $\rightarrow$ young, rich in metals
 $\downarrow$
everything
except H and He
 $\rightarrow$ found in spiral arms

population 2 $\rightarrow$ older metal-poor
found in bulge & halo



top view



new stars are usually found
at the outskirts of the spiral
arm (like renting house)

old stars at bulge.

population 3 → hypothetical, first generation stars

→ formed from pure H & He after the Big Bang

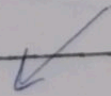
Stellar cloud → main sequence



Red giant



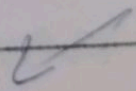
white dwarf



Black hole



neutron star



Supernova explosion

Life cycle of stars

Milky way galaxy

→ Ancient Mergers

→ Gaia Enceladus / Sausage

N - rich stars - rare in Milky way
- high probability it
is in a globular
cluster

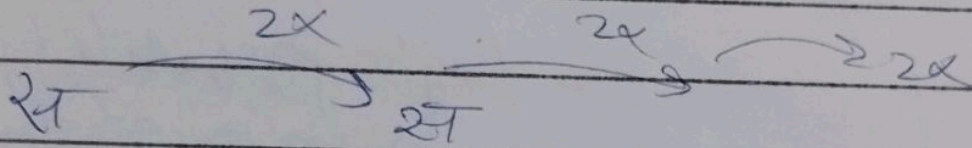
Ad Astra per aspera

a little excess of matter over
antimatter

matter + antimatter =
Energy

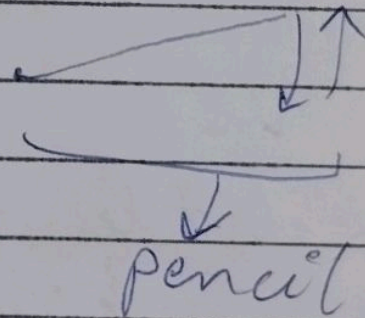
Quantum fluctuations

~~Vibration is motion~~



frequency

Mode shape



→ in sync - in phase

↓
to increase frequency

→ Rubber in bridges (metro bridges)

electromagnetic shaker for parts of rockets - test how they behave.

seismic centres